

Section 11. Multiple Random Variables

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Two Random Variables: Bayes

Bayes' rule: A useful formula to reverse the order of the conditionals

- ▶ For discrete r.v.'s X and Y

$$p_{Y|X}(y|x) = \frac{p_{XY}(x, y)}{p_X(x)} = \frac{p_{X|Y}(x|y) p_Y(y)}{\sum_{y \in \mathcal{Y}} p_{X|Y}(x|y) p_Y(y)}$$

- ▶ For continuous r.v.'s

$$f_{Y|X}(y|x) = \frac{f_{XY}(x, y)}{f_X(x)} = \frac{f_{X|Y}(x|y) f_Y(y)}{\int_{-\infty}^{\infty} f_{X|Y}(x|y) f_Y(y) dy}$$

Multiple Random Variables

Given a collection of random variables $X_1, \dots, X_n$, we can define the *random vector*

$$\mathbf{X} = \begin{bmatrix} X_1 & X_2 & \cdots & X_n \end{bmatrix}^T \in \mathbb{R}^n$$

- ▶ The *mean* of the random vector is

$$\mathbb{E}[\mathbf{X}] = \mathbb{E} \begin{bmatrix} X_1 & X_2 & \cdots & X_n \end{bmatrix}^T = \begin{bmatrix} \mathbb{E}[X_1] & \mathbb{E}[X_2] & \cdots & \mathbb{E}[X_n] \end{bmatrix}^T$$

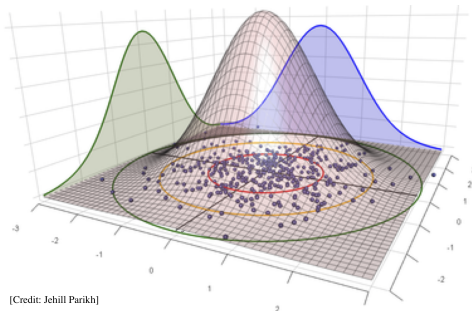
- ▶ The *covariance matrix* of the random vector is

$$\Sigma = \begin{bmatrix} \text{Cov}[X_1, X_1] & \cdots & \text{Cov}[X_1, X_n] \\ \vdots & \ddots & \vdots \\ \text{Cov}[X_n, X_1] & \cdots & \text{Cov}[X_n, X_n] \end{bmatrix}$$

Multiple Random Variables (cont.)

An important PDF is the multivariate Gaussian with mean $\mu \in \mathbb{R}^n$ and covariance $\Sigma \in \mathbb{R}^{n \times n}$, denoted as $\mathbf{X} \sim \mathcal{N}(\mu, \Sigma)$, defined as

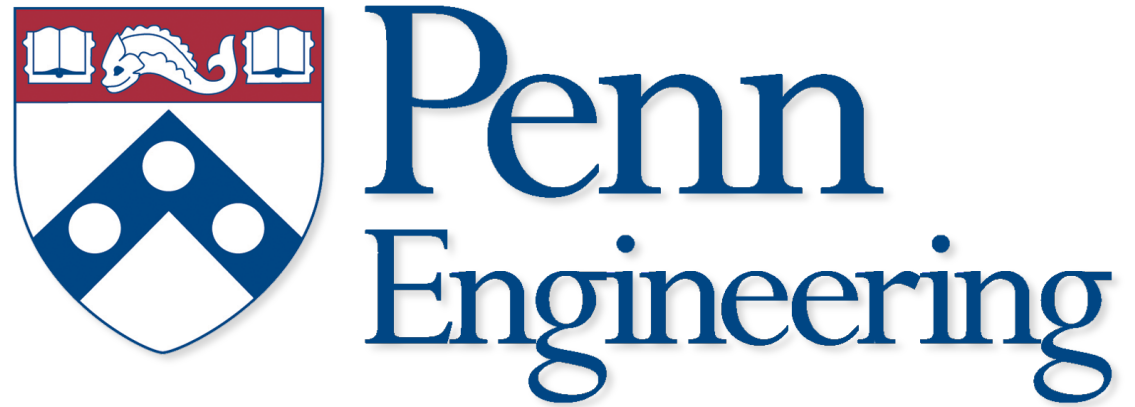
$$f_{\mathbf{X}}(\mathbf{x}; \mu, \Sigma) = \frac{1}{(2\pi)^{n/2} \det(\Sigma)^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu)^{\top} \Sigma^{-1}(\mathbf{x} - \mu)\right)$$



[Credit: Jehill Parikh]

Reading and next section:

- ▶ Reading: “*Review of Probability Theory*” by A. Maleki and T. Do
Available online at <<http://cs229.stanford.edu/section/cs229-prob.pdf>>
- ▶ Next section: Introduction to Statistical Learning...



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